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ORIGINAL CONTRIBUTION

Supervised Machine Learning for Isomorphic Identification of Epicyclic Gear Trains

Jiyaul Mustafa[1] · Shahnawaz Ahmad[2] · Nagendra Kumar Maurya[3] · Mohammed Wasid[5] · Vineet Kumar[4]

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Abstract
Isomorphism identification in epicyclic gear trains (EGTs) plays a crucial role for the structural synthesis and analysis of geared mechanisms since last two decades. Although many researchers have been suggested traditional approaches based on graph theory and matrix operations often face computational inefficiencies in handling complex gear structures. This paper introduced an automated approach for detecting isomorphism in polyhedral representations of EGTs using supervised machine learning. A curated dataset comprising various classes of conventional EGT graphs is used for training and classification through Graph Neural Networks (GNNs). To validate the classification results, the Total Distance Technique (TDT) is employed, wherein the TDT value corresponds to the sum of all elements in the adjacency matrix. Two EGTs are classified as isomorphic if both their GNN embeddings and TDT values align; otherwise, they are considered distinct. All results are fully satisfied with existing literature and the proposed approach is best applicable for optimising the design of mechanical transmission systems through machine learning-driven automation.

Keywords Epicyclic gear trains · Isomorphism · Supervised machine learning · Graph neural networks · Structural topology

Introduction

Compact design criteria and high-torque transmission capabilities have made epicyclic gear trains (EGTs) a widespread choice for automotive, aerospace, and industrial applications. However, the identification of isomorphic structures in EGTs continues to be one of the most important problems in kinematic synthesis. EGT isomorphism detection is traditionally based on graph theory, or sometimes in combination with adjacency matrices use as a surrogate, but they involve a lot of computation and are often ineffective. The application of isomorphic structure identification with machine-learning techniques may maximize the spent time for analysis and increase precision at the same time. In this work, EGT isomorphism detection with supervised and unsupervised learning models is investigated together with the integration of the two approaches, showing that data-centric methods are more effective than traditional ones in gear mechanism design.

Several studies have been done on the isomorphism detection problem in EGTs. For the most part, these methods are based on graph theory, the usage of adjacency matrices, and the path matrix approach to separate structurally identical mechanisms. Ravisankar and Mruthyunjaya [1] pioneered the development of computerized methods for structural analysis of kinematic chains synthesis. A novel method [2] using Wiener number for isomorphism detection in epicyclic geared mechanisms and planar kinematic chains set forth by Mustafa & Hasan for great years old duplicity problems. Jiyaul et al. [3] have proposed a non-redundant,

✉ Jiyaul Mustafa
jiyaul.alld@gmail.com

[1] Department of Mechanical Engineering, SEAS, Bennett University (The Times Group), 201310 Greater Noida, India

[2] Department of Computer Science Engineering and Technology, Bennett University (The Times Group), 201310 Greater Noida, India

[3] Department of Mechanical Engineering, GL Bajaj Institute of Technology & Management, 201308 Greater Noida, India

[4] Department of Mechanical Engineering, Sharda University, 201308 Greater Noida, India

[5] Department of Computer Science and Engineering, The LNM Institute of Information Technology, Jaipur, Rajasthan 302031, India

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graph theory-based construction of one degree of freedom epicyclic gear trains by means of adjacency matrices and wiener number. Peng [4] outlined a new way of designing speed ration gears in PGTs with variable degree of freedom, improving design functionality. Pennestri et al. [5] developed automatic algorithms for geared mechanisms static and dynamic analysis based on graph theory specialties. Chen and Chen [6] introduced a stable model for efficiency calculation in EGTs, closely aligning theoretical values with experimental data. Ding et al. [7] synthesized 1-DOF EGTs up to nine links using graph representation and characteristic number strings for isomorphism identification. Saini and Singh [8] and Rajasri et al. [9] employed hamming number techniques for isomorphism detection in KCs and EGTs, streamlining graphical analysis and synthesis processes. Mustafa et al. [10] have compared modified path matrix, Modified gradient, and Bocher's technique for isomorphism in epicyclic Gear Trains, assessing reliability, computational efficiency, and structural properties. Mustafa et al. [11] introduced a modified path matrix approach for isomorphism detection in epicyclic gear trains, addressing challenges in existing methods, validated through examples. Chu and Zou [12] provided insights into structural synthesis and topological graphs of planar multiple joint and geared kinematic chains. Yang et al. [13] proposed a novel approach for detecting rotational graphs and canonical rotational graphs of PGTs, introducing concepts of graph representation to avoid pseudo-isomorphism. Gao and Hu [14] delved into the topological synthesis and kinematic analysis of planetary transmission systems using graph theory, highlighting the relationship between speed ratios based on topology. Kamesh et al. [15] proposed an algorithm based on the net distance method for quantitative analysis of isomorphism in KCs and EGTs with 1-2 degrees of freedom (DOF), offering a straightforward calculation for isomorphism detection. Kamesh and others were able to establish parameters for the construction of EGTs enabling the detection of displacement and rotational graphs due to his vertex incidence polynomial theory [16]. The novel Isomorphism Modified Gradient Method (MGM) introduced by Mustafa provides a new efficient epicyclic gear train isomorphism detection application that has been validated [17]. Kamesh and others were able to establish parameters for the construction of EGTs enabling the detection of displacement and rotational graphs due to his vertex incidence polynomial theory [18]. Esmail [19] performed a complete kinematic analysis of PGTs using kinematics graph theory in an unconventional way by using it for determining speed ratios of multiple link mechanisms. Bal et al. [20] recommended the use of tree graphs for structural synthesis of PGTs which breaks away from the idea of using adjacency matrix and characteristic polynomial as the only alternatives. Shukla et al. [21], for example, used the gradient method to formulate an algorithm for isomorphism checking in KCs. The author proposes the concept of gradient strings for synthesis and analysis. Souza et al. [22] attempted the problem of generating kinematic chains of zero variety in EGTs intended for systems with one or two degrees of freedom. Rao et al. [23] put forward fuzzy entropy for grading epicyclic gear trains which contributed to systems of a greater number of machines. Yang and Ding [24] developed their own automated method for perimeter loop are the most simplistic for detection of isomorphism in the PGTs, which simplifies the process of eliminating duplicity. Mustafa et al. [25] utilized Bocher's equation for the purpose of isomorphism recognition in epicyclic gear trains, allowing for a novel way to address the problem. Liu et al. [26] suggested the idea of using a GA model in planar kinematic mechanisms for improved isomorphism recognition accuracy, making the process more efficient. Yang and Ding [27] researched on automating the detection of degenerate structures in EGTs which is important for a vehicle transmission system and in the field of aeronautics. Application and synthesis of automotive parts of engines are given in [28–30] for structural study of mechanisms. This paper describes an automated approach based on machine learning that enhances the design and analysis processes of EGTs structures using Graph Neural Networks (GNNs) to classify conventional different graph representations. TDT works by validating these results through the summation of elements of the adjacency matrix. EGTs are isomorphic if GNN embeddings TDT values are equal; otherwise, GNN embeddings are not equal, guaranteeing accurate and efficient results in topological analysis.

Graph Theory

Isomorphism detection and topological analysis within EGTs relies heavily on graph theory. Using matrix representations and graph algorithms, graph theory systematically identifies entire collections of structurally identical configurations, enhances design efficiency, and preserves kinematic integrity by modeling the linkages, joints, and mobility constraints using adjacency matrices.

Adjacency Matrix

The adjacency matrix shows the connection between two vertices or links in EGTs. The matrix demonstrates symmetry while all its diagonal positions contain zeros. The matrix derives from its functional schematic and shows as:

$$[A] = \{a_{ij}\} \tag{1}$$

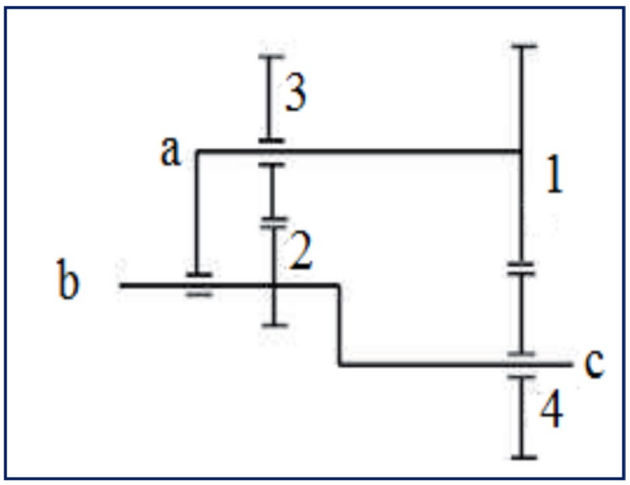


Fig. 1 Functional Schematic of four-links One-DOF EGTs

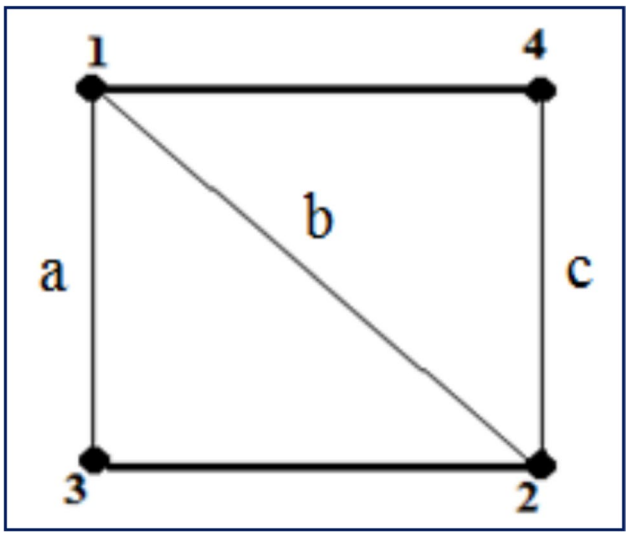


Fig. 2 Graph of four-links One-DOF EGTs (G-4001)

The formula uses (i, j) to indicate the row-column positions of the elements from matrix [A]. A process has been established to establish each element value in the matrix according to the provided below chart Fig. 1.

For n × n adjacency matrix can be written as,

$$[\mathbf{A}] = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \dots & \dots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \dots & \dots & a_{2n} \\ a_{31} & a_{32} & a_{33} & \dots & \dots & a_{3n} \\ \dots & \dots & \dots & \dots & \dots & \dots \\ \dots & \dots & \dots & \dots & \dots & \dots \\ a_{n1} & a_{n2} & a_{n3} & \dots & \dots & a_{nn} \end{bmatrix} \tag{2}$$

The adjacency matrix for four-link 1-DOF EGTs (Fig. 2) takes its form based on the given chart.

$$\text{AdjacencyMatrix, [Am]} = \{\text{a4} \times 4\} = \begin{bmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 \\ 1 & 2 & 0 & 3 \\ 2 & 1 & 3 & 0 \end{bmatrix} \tag{3}$$

Total Distance Technique

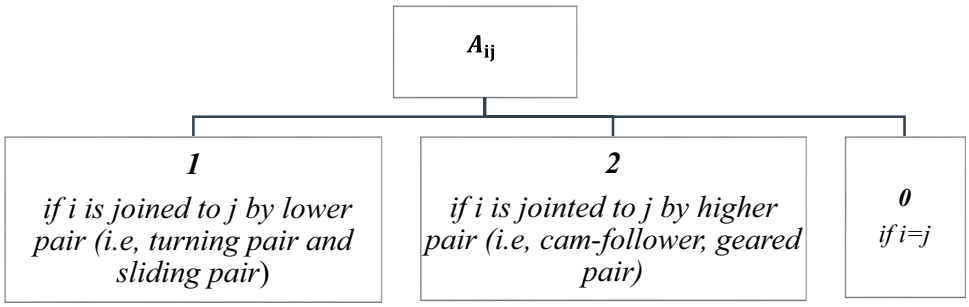
The "Total distance technique (TDT)" is topological approach to evaluate isomorphic property in the mechanisms. It can evaluate as,

$$\text{TDT} = \sum_{\text{i=1}}^{\text{n}} \sum_{\text{j=1}}^{\text{n}} (\text{A}) \tag{4}$$

where $\text{A} = \{\text{a}_{\text{ij}}\} = \text{Adjacency Matrixofaperticularconventionalgraph.}$

For Fig. 2, Value of TDT by using equation (4),

$$TDT = \begin{bmatrix} 4 \\ 4 \\ 6 \\ 6 \end{bmatrix} = 20$$



Machine Learning Approaches for Structural Analysis of EGTs

Machine learning based approach is a novel method to detect isomorphism in EGTs. There are seven techniques where detection of graphs of EGTs can be evaluated. Therefore, in this paper, we have used Graph Neural Networks (GNNs) for structural analysis of EGTs and their derived mechanisms.

(a) Graph Neural Networks (GNNs)

GNNs analyses EGT structures by learning graph embeddings, preserving connectivity and topological relationships. They classify isomorphic and non-isomorphic configurations by applying message passing, ensuring accurate detection and structural analysis of complex gear train mechanisms.

(b) Convolutional Neural Networks (CNNs) on Graph Representations

CNNs process graph-based images of EGTs, capturing structural patterns through feature extraction and convolutional layers. This method allows for the proper automation of structural analysis for the classification of gear mechanisms and the verification of their optimal designs.

(c) SVMs For Topological Classification

SVMs differentiate between configurations of EGT structures with topological features that have been specifically designed for them, and with EGT structures parameters that have been set. Its capability to detect isomorphic graphs is very efficient and assists in systematic synthesis and optimization of gears.

(d) Decision Trees and Structural Recognition with Random Forests

Decision trees are utilized to analyze EGT for their topology through connectivity and kinematic parameters. Those models effectively classify and analyze gear mechanisms with the objective of optimizing time spent and accuracy used in design for manual analysis.

(e) Autoencoders For Feature Learning and Dimensionality Reduction

Autoencoders create representations of EGT structures with low complexity and high dimensionality while preserving important structural details. This technique minimizes the effort required for computation in analyzing gear trains which in turn raises the classifying capability of machine learning models.

(f) Gear System Optimization with Reinforcement Learning

In reinforcement learning, an agent is taught to alter the configuration of the gears using the received rewards as motivation. It innovatively searches through the most effective topologies that ensure good mechanical transmission of power and enhances the design of the train of gears for improved efficiency and durability.

(g) K-Means Clustering for Structural Grouping

EGT structures by means of graph EGT structures with isomorphic components are grouped together and are automatically added into a certain K-Measure cluster. This ensures easier automated classification of gear topologies which enhances the comparison of structures and mechanical designs standardization.

Proposed Methodology: GNN for Isomorphism Detection in EGTs

The detection of isomorphism is critical for removing redundancies in the design of a mechanism. This paper presents a machine learning technique where Graph Neural Networks (GNNs) are used to analyze EGT graphs. GNNs are selected because epicyclic gear trains can be naturally represented as graphs, where nodes correspond to links and edges represent joints. GNNs are well suited for learning structural relationships directly from graph-structured data while preserving topological connectivity, which is essential for identifying isomorphic structures.

Compared with conventional machine learning methods such as CNNs, SVMs, and decision trees, GNNs can directly process graph data without flattening features and effectively capture relationships between neighboring nodes through message-passing mechanisms. This ability to retain topological information makes GNNs more effective and accurate for graph classification tasks, particularly for detecting structural similarities among epicyclic gear train configurations.

The following steps and methodology ensure accurate isomorphism detection and structural analysis.

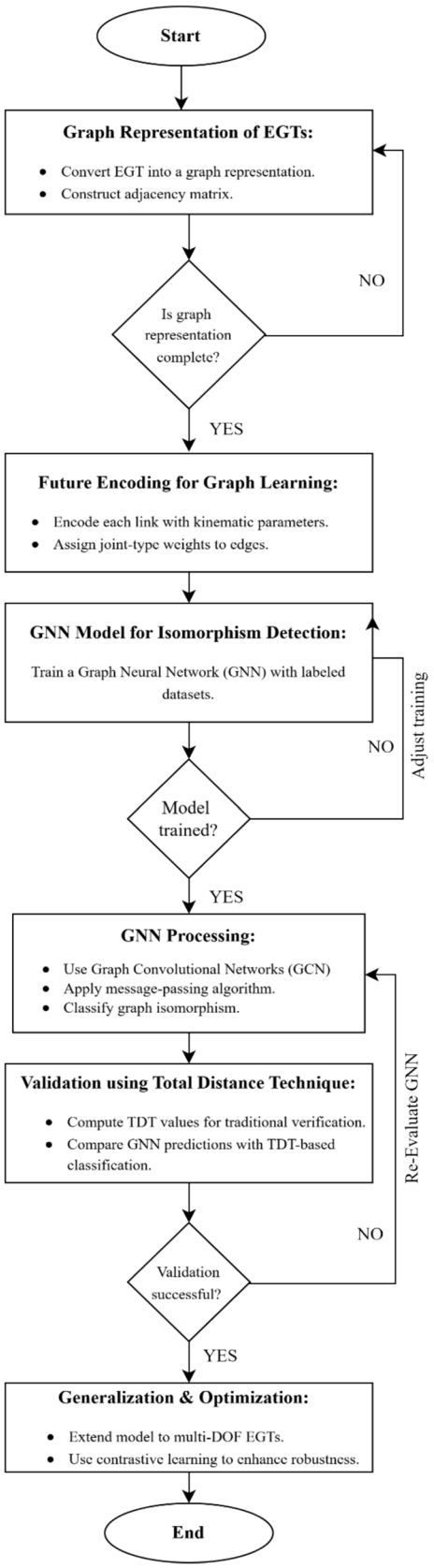


Fig. 3 Flow chart for analysis of isomorphic properties in EGTs by GNN technique

a. Graph Representation of EGTs

- Convert each EGT into a graph where nodes represent links and edges represent joints (turning or sliding pairs).
- Construct the adjacency matrix from this representation.

b. Feature Encoding for Graph Learning

- Each link (node) is encoded with kinematic parameters (e.g., gear ratio, DOF, type of connection).
- Edges carry joint-type weights to capture kinematic constraints.

c. GNN Model for Isomorphism Detection

- Train a Graph Neural Network (GNN) using labeled datasets of known isomorphic and non-isomorphic EGTs.
- Use Graph Convolutional Networks (GCN) or Graph Isomorphism Networks (GIN) to learn structural similarities.
- Apply a message-passing algorithm to aggregate neighborhood features and classify whether two graphs are isomorphic.

d. Validation using Total Distance Technique (TDT)

- Compute TDT values as a benchmark for traditional verification.
- Compare GNN-predicted results with TDT-based classification for accuracy validation.

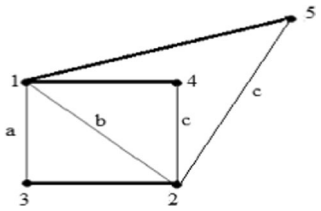
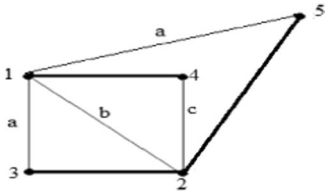
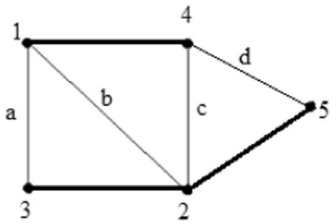
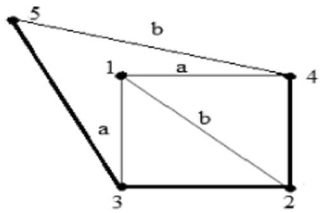
e. Generalization & Optimization

- Extend the model to multi-DOF EGTs mechanisms.
- Optimize the model using contrastive learning to improve robustness against minor variations in topology.

Fig. 3 shows the complete flow chart for analysis of isomorphic properties in EGTs by GNN technique.

The datasets have created from previously reported EGT structures available in literature related to graph-theoretic synthesis of planetary gear trains and epicyclic mechanisms. These include works by Ravishankar and Mruthyunjaya [1], Ding et al. [24], Yang and Ding [27], Souza et al. [22] and Tsai [31, 32]. The adjacency matrices of these structures

Table 1 Topological synthesis of 4 graphs 5-links 1-DOF epicyclic gear trains

Graph no	Kinematic graph	Adjacency matrix	Degree sequences (GNN model)	TDT
EGT-5001		$\begin{bmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 1 & 1 \\ 1 & 2 & 0 & 3 & 3 \\ 2 & 1 & 3 & 0 & 2 \\ 2 & 1 & 3 & 2 & 0 \end{bmatrix}$	[6 5 9 8 8]	38
EGT-5002		$\begin{bmatrix} 0 & 1 & 1 & 2 & 1 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 3 & 2 \\ 2 & 1 & 3 & 0 & 3 \\ 1 & 2 & 2 & 3 & 0 \end{bmatrix}$	[6 5 9 8 8]	38
EGT-5003		$\begin{bmatrix} 0 & 1 & 1 & 2 & 3 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 3 & 4 \\ 2 & 1 & 3 & 0 & 1 \\ 3 & 2 & 4 & 1 & 0 \end{bmatrix}$	[7 6 10 7 10]	40
EGT-5004		$\begin{bmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 & 3 \\ 1 & 2 & 0 & 2 & 2 \\ 1 & 2 & 2 & 0 & 1 \\ 2 & 3 & 2 & 1 & 0 \end{bmatrix}$	[5 8 7 6 8]	34

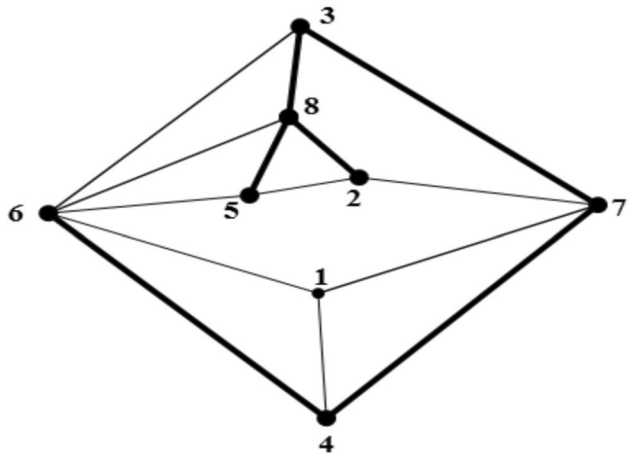


Fig. 4 Kinematic Graph of 8-links EGT-8001

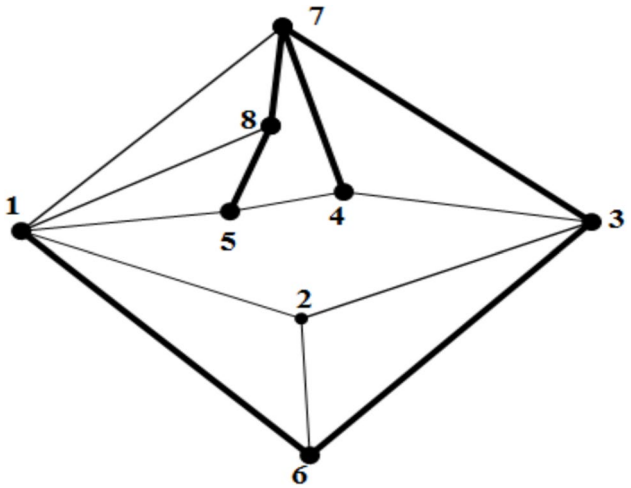


Fig. 5 Kinematic Graph of 8-links EGT-802

were used as graph inputs to train the GNN model. Thus, the dataset is constructed from published literature and synthesized graph representations of EGTs with different numbers of links and degrees of freedom.

Results & Discussion

The proposed method based on supervised machine learning is applied on different graphs of five-links single degree of freedom EGTs to identifying the isomorphism and

Table 2 Topological synthesis of two graphs 8-links 1-DOF epicyclic gear trains

Graph No	Adjacency matrix	Degree sequences (GNN model)	TDT
EGT-8001	$\begin{bmatrix} 0 & 2 & 3 & 1 & 2 & 1 & 1 & 4 \\ 2 & 0 & 3 & 3 & 1 & 2 & 1 & 2 \\ 3 & 3 & 0 & 4 & 2 & 2 & 1 & 2 \\ 1 & 3 & 4 & 0 & 3 & 2 & 2 & 2 \\ 2 & 1 & 2 & 3 & 0 & 1 & 2 & 2 \\ 1 & 2 & 2 & 2 & 1 & 0 & 2 & 1 \\ 1 & 1 & 1 & 2 & 2 & 2 & 0 & 3 \\ 4 & 2 & 2 & 2 & 2 & 1 & 3 & 0 \end{bmatrix}$	$\begin{bmatrix} 14 \\ 14 \\ 17 \\ 17 \\ 13 \\ 11 \\ 12 \\ 16 \end{bmatrix}$	114
EGT-8002	$\begin{bmatrix} 0 & 1 & 2 & 2 & 1 & 2 & 1 & 1 \\ 1 & 0 & 1 & 2 & 2 & 1 & 3 & 2 \\ 2 & 1 & 0 & 1 & 2 & 2 & 2 & 4 \\ 2 & 2 & 1 & 0 & 1 & 3 & 2 & 3 \\ 1 & 2 & 2 & 1 & 0 & 3 & 2 & 2 \\ 2 & 1 & 2 & 3 & 3 & 0 & 3 & 3 \\ 1 & 3 & 2 & 2 & 2 & 3 & 0 & 2 \\ 1 & 2 & 4 & 3 & 2 & 3 & 2 & 0 \end{bmatrix}$	$\begin{bmatrix} 10 \\ 12 \\ 14 \\ 14 \\ 13 \\ 17 \\ 15 \\ 17 \end{bmatrix}$	112

Table 3 Generation of epicyclic gear trains up to 10 Links with 1—2 DOFs

Sr. No	No. of links	Number of DOF	Total number of generated graphs	Used graphs for isomorphism	Discarded graphs	Duplicated graphs
1	4	1	2	2	0	0
2	5	1	6	6	0	0
3	6	1	26	26	0	0
4	6	2	50	35	10	5
5	7	1	124	124	0	0
6	7	2	200	150	30	20
7	8	1	600	500	50	50
8	8	2	1000	800	100	100
9	9	1	3000	2500	300	200
10	9	2	5000	4000	600	400
11	10	1	15,000	12,000	2000	1000
12	10	2	25,000	20,000	3000	2000

synthesis analysis. Table 1 shows the values of TDT and degree sequences of graph EGT-5001 & EGT-5002 are identical so these graphs are isomorphic in nature. Whereas values of TDT and degree sequences of graph EGT-5003 & EGT-5004 are not similar so these graphs are structurally distinct in nature.

In Table 1, The term EGT-5001 represents a specific graph configuration of a five-link single-degree-of-freedom epicyclic gear train. In this notation, EGT denotes Epicyclic Gear Train, 5 indicates the number of links, and 001 represents the unique configuration number within the generated graph set.

The applications of two kinematic graphs of 8-link, 1-DOF epicyclic gear trains are presented in Fig. 4 and Fig. 5. The topological synthesis results are given in Table 2. The degree sequences and TDT values of the two graphs are different, indicating that they are non-isomorphic in nature.

This technique is then applied to generation of EGTs graphs up to 10 links and 1-2 DOF system. Table 3 shows the complete results of epicyclic gear train graphs upto 10 links by supervised machine learning technique.

Conclusion

This study presents a novel approach for detecting isomorphism in epicyclic gear trains (EGTs) using supervised machine learning, specifically Graph Neural Networks (GNNs), integrated with the Total Distance Technique (TDT). Traditional graph theory-based methods, such as adjacency matrices and characteristic polynomials, often face computational inefficiencies when analyzing complex EGT structures. The proposed GNN-based methodology overcomes these limitations by learning graph embeddings that capture kinematic and topological features, enabling accurate classification of isomorphic and non-isomorphic EGT configurations. The TDT validates these results by computing the sum of adjacency matrix elements, ensuring alignment with GNN predictions. Applied to EGTs with up to 10 links and 1-2 degrees of freedom, the approach successfully identifies isomorphic graphs, eliminates redundancies, and detects degenerate structures, as demonstrated in the topological synthesis of five-link 1-DOF EGTs and extended to higher-link systems. The integration of GNNs

with TDT enhances computational efficiency and precision, offering a scalable solution for structural synthesis and kinematic analysis. This machine learning-driven automation optimizes the design of mechanical transmission systems, with potential applications in automotive, aerospace, and industrial sectors, paving the way for future advancements in intelligent gear mechanism design.

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Declarations

Conflict of Interest The author states they did not receive money from competing sources or face situations that may have changed the research shown in this report.

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